O_3 toxicity in plants has been studied by Elstner *et al*. (1991)[7]; Batini *et al*. (1995)[8]; Pitcher *et al*. (1996)[9]; Polle *et al*. (1995)[10]; Melhorn *et al*. (1994)[11]; Menser (1964)[12]; Castillo *et al*. (1987)[13]; Chameides (1989)[14]; Heritt *et al*. (1990)[15]; Kangasjarvi *et al*. (1994)[16]; Anttonen *et al*. (1996)[17]; Schraudner *et al*. (1996)[18]; Sandermann (1996)[19].

Like any other gas, O_3 enters the plant leaves through stomata as part of normal atmospheric gas exchange, goes in apoplast, it reacts quickly with biomolecules, water located there. Some of it, may be converted into secondary ROS (Reactive Oxygen Species) such as superoxide anion radicals ($\dot{o}_2$), hydroxy radicals ($\dot{o}H$), H_2O_2 and singlet oxygen – Hoigne *et al*. (1975)[20]; Koppenol (1982)[21]; Grimes *et al*. (1983)[22]; Byvoel *et al*. (1975)[23]; Kanofsky *et al*. (1995)[24]; Extremely higher dose of O_3 permeate to the underlying plasma membrane. O_3 reacts with apoplastic bimolecules & generate toxic species (lipid peroxidation products) that initiates damage to plasma membrane lipids & proteins resulting into decreased photosynthesis, electrolyte leakage and accelerated senescence.

Heat wave / heat stress - labour hours lost

In India 40 million heat wave exposure cases have been reported compared to 2012 there by the country lost nearly 75,000 million hours of labour in 2017 (equivalent to a year's work for 7% of working population) compared to about 43,000 million hours in 2000, an increase of more than 30 million labour hours lost in less than two decades.

Forest Fires

Dry and deciduous forests in the borderlands of Chattishgarh, Maharashtra & Telangana are affected by fires every year. Parts of western Himalayas such as Uttarakhand & Himachal Pradesh experienced severe forest fires. "Forest fires season" is time after winters when there are natural changes of forests catching fire due to rise in temperature.

Forest fires were also witnessed in Uttarakhand (Pauri, Garhwal, Rudra prayag, Nainital, Chamoli, Udham Singh Nagar, Bageshwar, Champawat & Pittoragarh. All this was reported by Roy (2018)[1]. According to the state forest department there have been 989 fire incidents between Oct. 1, 2020 & April 4, 2021 in Uttarakhand, destroying around 1,297.43 hectares of rich biodiverse forests. Since 2000 forest fires have affected over 48,000 hectares in the said state- report state forest. The causative factors are anthropogenic, carelessness & due to rise in temperature less of 'rain in winter etc.

REMEDIAL MEASURE

In order to combat increase in temperature strong international co-operation is needed to stop / minimize the causative factors to a level which does not destroy the environment. Following are advisable

1. Shift to low or zero emission
2. Electrifying transport
3. Development of green infra - structure
4. Increasing the energy efficiency.

Chandra Bhusan Deputy Director General of Centre for Science and Environment (CES) opines that "through it will be very difficult in the current global economic system to limit warming to 1.5°C, it is not impossible. This will require acting on all fronts to rapidly reduce emission by 2030 without an active participation of the United States, this will be impossible".

The present authors suggest that "plants with more CO2 absorbance capacity can be developed by 'gene editing technology' (CRISPR technology - i.e. clustered regularly interspersed short palindromic repeats".

The idea may appears to be fantastic yet an effort in the direction needs consideration.

There are trees available which imbibe air pollution which can be planted as for needs. Plants have to be selected on the basis of the types of pollutants, their intercity, location and easy availability & suitability of the climate conditions. Hydrogen is considered a source of clean energy. As a fuel it can run electric motos, or burned in internal combustion engine & has a number of industrial uses. It can be compressed or converted to liquid, easily stored.

Normally, hydrogen is produced through stream methane reforming and coal gasification. This raised the fuel's carbon foot print, despite it burns cleanly. The traditional catalysts through produce hydrogen from water yet it relies on expensive platinum group elements raising the cost and therefore large scale splitting becomes rather impractical.

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Isolation and screening of potassium solubilizing soil bacteria from Rhizosphere of paddy plant

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Abstract- Potassium solubilizing bacteria were isolated from the rhizosphere of the Paddy plant. Altogether seven isolates were collected from the 15 villages of Madhepura district. All isolates were cultured in Aleksandrow medium and colonies were further sub-cultured in nutrient medium. Morphological and biochemical test were performed for colony morphology, gram-staining, Citrate-utilization test, Voges-Proskauer test, Catalase test, Oxidase test and Methyl test were performed. For the determination of solubilizinig efficiency and solubilisation index bacteria were cultured in Aleksandrow medium, Halozone and colony diameter were measured and Solubilizing Efficiency (SE) and Solubilization Index (SI) were calculated. The highest value was observed in isolate no. K_1 (SE=11.33, SI=4.581) and the minimum for isolate no. K_4 (SE=5.66, SI=2.698).

Key words: Solubilization Index, Solubilizing Efficiency, Citrate test, Catalase test, Oxidase test

INTRODUCTION

Potassium is an essential macronutrient for plants. Among the three essential nutrients required by the plant one of them is Potassium. It plays a vital role in enzyme activation, osmotic regulation, opening and closing of stomata, starch synthesis, Protein synthesis and water nutrient transport. Soil contain large amount of potassium, however most of the Potassium is unavailable for plant uptake. Application of chemical fertiliser is one of the sources to fulfil potassium requirement of plants but chemical fertilizers have a considerable negative effect on soil. The eco-friendly agriculture system has emerged as an important thrust area globally for long-term soil environmental sustainability and to minimise the environmental pollution associated with extensive use of chemical fertiliser to conserve our existing resources and to minimise environmental pollution hazards.[1-5] Hence, the production and management of bio fertiliser containing KSB can be an effective alternative to chemical fertiliser. It is well-known that potassium solubilizing bacteria *Bacillus subtilis*, *Bacillus mucilaginous*, *Bacillus megaterium*, *Paenibacillus macerans*, *Paenibacillus agglomerans*, *Pseudomonas aureofaciens*, *Pseudomonas fluorescens*, *Agrobacterium radiobacter*, *Azospirillum brasilense*, *Azospirillum lipoferum*, *Azotobacter chroococcum* etc. can solubilize Potassium which become available to Plants.

These soil bacteria are commonly known as KSB (Potassium Solubilizing Bacteria). The diversity and ability of KBS depends upon soil and climatic condition. KSB are able to dissolve silicate minerals and release K through the production of organic and inorganic acids,

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Acidolyses, Polysaccharides, Complexolysis, Chelation and exchange reactions.

MATERIAL & METHOD

The soil samples were collected from rhizosphere of Paddy plants from different fields. Each sample consisted of 200 g of soil. Samples were collected in septic bags and transported to lab for further processing. Details of soil sample and culture media used are mentioned in Table no. 01.

Isolation of Potassium solubilizing bacteria:

1 gm of soil sample was mixed thoroughly in 10 ML autoclaved distilled water. Serial dilution was made up to 10^{-6}. Diluted soil samples were placed on Aleksandrow medium. The plates were incubated at 30°C for three days. Colonies exhibiting clear zones were selected as potassium solubilizer. Selected colonies were placed on nutrient agar medium for obtaining pure culture. The pure culture was again inoculated in Aleksandrow medium to confirm their K solubilizing efficiency.

Estimation of Solubilisation index and Solubilizing efficiency:

Qualitative analysis of K solubilizing activity of various isolates was calculated by measuring halozone and colony diameter on Aleksandrow medium. Halozone and colony diameter was measured using Calipers.

Calculation of Solubilization Efficiency (SE) and Solubilization Index (SI) were calculated by following formula:

$$SE = \frac{\text{Halozone} - \text{Colony diameter}}{2}$$

$$SI = \frac{\text{Halozone}}{\text{Colony diameter}}$$

RESULT

A total of 15 soil samples were collected from different Paddy fields of following villages of Madhepura district: Rampatti, Chousa, Budhma, Murliganj, Barbana. Details of soil sample, Date of collection, Culture medium and Isolate code are described in Table no. 01. All together seven isolates were selected as K solubilizing bacteria for morphological and biochemical characterisation. All isolates were examined for colony morphology, gram-staining, Methyl red test, Voges-Proskaurtest, Nitrate reaction test, Catalase test and Citrate test. The morphological characters are tabulated in Table no. 2. Biochemical characters are tabulated in Table no. 3. The value of Potassium solubilizing efficiency and Potassium solubilization index are mentioned in Table no. 4.

Table 1- Details of K-solubilizing Bacteria isolated from different soil samples

S. No.	Soil Sample	Place	Date	Culture medium	Isolate code
1	S_1	Rampatti	03-08-20	NA-AM	K_6
2	S_2	Barbana	05-08-20	NA-AM	K_2
3	S_3	Chousa	09-08-20	NA-AM	K_4
4	S_4	Rampatti	12-08-20	NA-AM	K_1
5	S_5	Rampatti	17-08-20	NA-AM	K_6
6	S_6	Budhma	22-08-20	NA-AM	K_5
7	S_7	Chousa	28-08-20	NA-AM	K_3
8	S_8	Barbana	05-09-20	NA-AM	K_1
9	S_9	Murliganj	08-09-20	NA-AM	K_4
10	S_{10}	Chousa	12-09-20	NA-AM	K_7
11	S_{11}	Barbana	18-09-20	NA-AM	K_4
12	S_{12}	Budhma	22-09-20	NA-AM	K_6
13	S_{13}	Murliganj	24-09-20	NA-AM	K_7
14	S_{14}	Budhma	26-09-20	NA-AM	K_1
15	S_{15}	Murliganj	29-09-20	NA-AM	K_2

NA: Nutrient-Agar medium
AM: Aleksandrow medium

Table 2- Morphological Character

Sl. No.	Isolate	Form	Elevation	Colony size	Colony colour	Cell shape	Gram staining	Remark
1	K_1	Regular	Flat	Small	White	Oval	-ve	*Cocci*
2	K_2	Regular	Flat	Small	White	Oval	-ve	*Cocci*
3	K_3	Irregular	Flat	Small	White	Rod	+ve	*Bacillus*
4	K_4	Irregular	Flat	Small	White	Oval	-ve	*Cocci*
5	K_5	Regular	Flat	Small	White	Oval	-ve	*Cocci*
6	K_6	Regular	Flat	Small	White	Oval	-ve	*Cocci*
7	K_7	Regular	Flat	Small	White	Oval	-ve	*Cocci*